Artificial Economies for Learning Agents: Testing Whether Institutions Stay Robust as Agent Capability Changes

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Draft v0.2 – July 13, 2026

Abstract

This paper introduces a code-first experimental platform for studying whether economic mechanisms remain robust when the strategic agents inside them are adaptive learners rather than equilibrium solvers. The platform implements a shared `World`/`Agent`/`Institution` interface, repeated multiseed evaluation, exploitability tests, and interchangeable mind classes ranging from random and tabular Q-learning agents to PyTorch DQN, PPO, independent-DQN, and centralized-critic agents. The first validated environment is a repeated two-firm pricing arena with six regulatory mechanisms. In this world, price caps robustly reduce exploitability across mind classes, but their effect on collusion is metric- and capability-dependent: stronger DQN-family learners can sit effectively at the cap while preserving high profit through the quantity margin. The second environment, Resource Island, extends the same interface to a spatial inventory economy and reveals a different lesson: institution conclusions are only meaningful when the world actually exercises the institutional channel. Early Resource Island versions mechanically under-tested property rights and trade price controls; a hardened v1 now creates contested resources, unequal trades, and specialization pressure. Three additional environments extend the ladder: Auction House provides a benchmarked auction-design testbed, Public Goods exposes common-pool extraction and contribution incentives, and Labor Market tests learned reporting under deferred acceptance. The current contribution is both empirical and methodological: it reports initial evidence that institutional robustness can change with learner capability, and it documents the validation discipline needed to distinguish real economic findings from inactive-mechanism artifacts.

First-Page Build Map

This draft is intentionally structured as the arXiv-facing version of the project, with the parts that already have evidence separated from the parts that still need final runs or prose.

Part	What It Must Do	Current Status
Motivation	State the core question: do mechanisms stay robust under a capability ladder of learning agents?	Ready to draft.
Platform	Define `World`, `Agent`, `Institution`, metrics, manifests, parity tests, and output validation.	Built and tested.
Pricing Arena	Main mature result: collusion, welfare, exploitability, and capability dependence in a repeated pricing game.	Full n=20 outputs exist; fixed independent-DQN is merged into the main table.
Resource Island	Non-pricing stress test: spatial resources, inventory, trade, property, redistribution, reputation.	v0 diagnosed; v1 and cross-mind full outputs validated.
Auction House	Third world: sealed-bid auctions with theory benchmarks.	Mechanics, benchmarks, Q-learning full run, P.6 smoke, and full P.6 queue are active.
Public Goods	Common-pool resource world with contribution, extraction, sanctions, matching, information restriction, reputation, and taxes.	Q-learning full run and P.6 smoke exist; full P.6 queue is active.
Labor Market	Two-sided matching world with learning workers and fixed employer preferences under deferred acceptance.	Q-learning full run and P.6 smoke exist; full P.6 queue is active.
Synthesis	Convert runs into thesis claims: which institutional effects generalize, which are world-specific, and which are capability-sensitive.	Preliminary five-world smoke synthesis exists; final claims wait on current full P.6 queues.
Limitations	Be explicit about inactive institutions, finite compute, simple worlds, and non-equilibrium learned behavior.	Ready to draft.

Stop condition for the first complete paper draft: after the current Auction House, Public Goods, and Labor Market full P.6 queues validate cleanly, freeze the build-out temporarily and write the cross-world synthesis before adding another environment or LLM-agent layer.

1 Introduction

Economic mechanisms are usually evaluated against a model of strategic behavior: equilibrium prices, truthful bidding, efficient allocation, or some other theoretical benchmark. That approach is powerful, but it leaves open a practical question that becomes more important as deployed systems are increasingly populated by adaptive software agents: what happens when the agents do not solve the theorist's equilibrium, but instead learn from repeated interaction?

This paper studies that question with artificial economies: small, controlled economic worlds in which institutions can be swapped while the learning architecture of the agents is varied. The goal is not to simulate a whole economy. The goal is to create environments that are simple enough to audit but rich enough to expose failures of institutional robustness. A mechanism is treated as robust only if it performs well across seeds, across learner classes, and under adversarial or exploitability-style checks.

The central research question is:

> When agent capability increases from random behavior to tabular Q-learning and deep reinforcement learning, do the same economic institutions still reduce collusion, exploitability, inequality, and inefficient strategic behavior?

The current evidence already gives a nontrivial answer in the pricing world. Price caps reduce exploitability for every tested mind class, but their collusion effect depends on the operational metric and on learner capability. In the full Pricing Arena comparison, tabular Q-learning under no regulation has average price 5.49, total profit 365.34, and exploitability 95.69. Under a price cap, average price falls to 4.73 and exploitability falls to 40.32. For DQN-family learners, however, the cap can create a different pattern: the agents sit near the ceiling, with low price dispersion, while preserving high profit-normalized collusion through quantity and demand effects. That is the kind of result this platform is meant to surface: a regulation can look strong under one behavioral lens and weaker under another.

2 Related Work

This project sits at the intersection of algorithmic collusion, reinforcement learning, and mechanism design. The Pricing Arena is motivated by work showing that learning algorithms can sustain supra-competitive outcomes in repeated pricing environments [Calvano et al., 2020, 2021]. The exploitability protocol is related to the broader idea that a learned policy should not only be evaluated by its on-policy average outcome, but also by its brittleness against strategic deviation.

The mind ladder follows standard reinforcement-learning baselines: tabular Q-learning and deep Q-networks [Sutton and Barto, 2018, Mnih et al., 2015], PPO [Schulman et al., 2017], and independent multi-agent Q-learning in the spirit of Tan [1993]. The Auction House environment connects the platform back to classic mechanism-design results: truthful bidding in second-price auctions [Vickrey, 1961], revenue and allocation tradeoffs in auction design [Myerson, 1981, Milgrom, 2004], and strategic bid shading in first-price auctions.

The Public Goods and Resource Island environments connect the platform to common-pool resource and social-dilemma settings, where individually profitable extraction can destroy shared surplus and where sanctions, reputation, and matching contributions are standard institutional responses. The Labor Market environment connects to two-sided matching and deferred acceptance, where truthful revelation and stability provide unusually sharp benchmark concepts.

[TODO: Tighten this section with exact citations for Calvano et al. 2021, Asker–Fershtman–Pakes, and Banchio–Skrzypacz once the literature pass is done.]

3 Platform

3.1 Interfaces

The codebase is organized around three abstractions:

- A `World` defines state transitions, observations, metrics, and rendering state.

- An `Agent` or mind maps observations to actions and updates from rewards and next observations.
- An `Institution` modifies rules, rewards, constraints, or information inside a world.

This separation is not cosmetic. It makes it possible to ask whether a result is about a world, an institution, or a learner. It also creates a regression test: if a new world requires rewriting the shared learning interface, then the platform abstraction has failed.

3.2 Validation Discipline

The platform treats a result as reportable only after three checks. First, the mechanics must be unit-tested: allocation, movement, payment, trade, collision, or reward rules must match hand-computed cases. Second, the experiment must be replicated over seeds, with per-seed and aggregate CSV outputs. Third, the institution must actually activate. If a property-rights rule never faces a non-owner gather attempt, or a trade price control can never bind because every trade is one-for-one, then the result is not evidence about that institution. It is evidence about the world design.

This distinction matters for Resource Island. The initial version ran and passed tests, but three institutions were mechanically under-exercised. Rather than report those as null effects, the project added diagnostic counters and a v1 pressure configuration with contested resources and unequal trades.

3.3 Mind Ladder

The current mind ladder includes random agents, tabular Q-learning, PyTorch DQN, PyTorch PPO, independent-DQN, and a centralized-critic scaffold. The DQN implementation uses replay, a target network, Huber TD loss, and a DeepMind-style centered RMSProp optimizer. PPO uses categorical actions, GAE, clipped policy loss, value loss, entropy regularization, and minibatch epochs.

An important implementation issue was discovered and fixed during the project: the original independent-DQN condition behaved like a DQN alias in several tables. The corrected implementation now uses separate per-agent learners, decorrelated child random streams, separate replay and exploration randomness, and scoped PyTorch initialization. Fixed independent-DQN replacement rows are now merged for Pricing Arena and Resource Island, and the same corrected condition is used in the active Auction House, Public Goods, and Labor Market P.6 queues.

4 Environment 1: Pricing Arena

4.1 World

Pricing Arena is a repeated two-firm pricing market. Each firm chooses a price from a discrete grid. Consumers choose between firms according to a noisy logit demand model. Firms earn profit from demand net of costs, and learning agents update from repeated play. The implemented mechanisms are no regulation, price cap, high-price tax, random audits, anti-collusion penalty, and demand shocks.

4.2 Metrics

The main metrics are average price, total profit, consumer surplus, welfare, exploitability, and two collusion indexes. The historical project metric is a price-normalized proxy. The literature-comparable metric is a profit-normalized index:

$$\mathrm{CI}_{profit} = \frac{\pi_{observed} - \pi_{Nash}}{\pi_{monopoly} - \pi_{Nash}}. \tag{1}$$

Both are reported because the divergence between them is empirically informative.

4.3 Main Result

Table 1 shows a compact version of the mature Pricing Arena result for Q-learning and random baselines. Price caps reduce exploitability and raise welfare relative to the no-regulation case in these rows.

Mind	Mechanism	Price	Profit	Welfare	Profit CI	Exploitability
Q-learning	none	5.49	365.34	810.43	0.475	95.69
Q-learning	price cap	4.73	342.42	829.99	0.424	40.32
Random	none	5.51	277.92	785.05	0.282	54.62
Random	price cap	4.32	261.89	819.83	0.247	22.84

Table 1: Selected Pricing Arena full-run results from the Phase 3 comparison output.

The more interesting finding appears when learner capability increases. In the Phase 3 table, price caps reduce exploitability across tested minds, but collusion is metric-sensitive for stronger learners. DQN-family agents can sit effectively at the cap, with low price dispersion, while maintaining high total profit. The current diagnosis is that the cap constrains the price margin but does not eliminate the quantity/profit channel.

[TODO: Add a full table comparing DQN, PPO, centralized-critic, and corrected independent-DQN under price cap versus no regulation.]

5 Environment 2: Resource Island

5.1 World

Resource Island is a bounded-grid gather/trade economy. Agents move, gather food and wood, maintain energy, accumulate inventories, and can die from starvation. The world contains property rights, redistribution, trade price controls, and reputation institutions. It was built to test whether the platform abstractions work outside a two-firm pricing game.

5.2 What v0 Taught

The first full Resource Island runs were mechanically valid but economically under-activated. In the initial full run, successful trade was sparse and some institutions did not bind. Follow-up replay showed that property rights almost never faced behavioral pressure: even when claims existed, non-owner opportunities to gather from claimed resource cells were extremely rare. Similarly, trade price controls could not bind because the original trade protocol only allowed one-for-one exchange.

This is an important negative result for the platform: a world can be runnable, tested, and statistically replicated while still failing to exercise the institution it is supposed to study.

5.3 v1 Hardening

Resource Island v1 adds four changes:

1. contested resource layouts, so multiple agents have reason to access the same cells;
2. unequal trade offers, so price controls can actually bind;
3. specialization pressure through heterogeneous resource preferences and starting inventories;
4. activation diagnostics for trade attempts, blocked trades, property opportunities, property violations, and institution blocks.

The v1 validation run verified that the economic channels were active, and the subsequent n=20 full run preserves that activation. Table 2 shows the medium validation values used as the pre-full-run gate.

Institution	Welfare	Survival	Trades	Trade Blocks	Property Opps.
none	1.095	0.933	6.034	0.000	0.000
property rights	1.078	0.923	5.790	0.000	17.734
trade price controls	0.875	0.898	0.000	5.779	0.000
reputation	1.313	0.927	6.867	0.000	0.000

Table 2: Resource Island v1 validation results.

The interpretation is straightforward. Price controls now bind because the world permits unequal trades. Reputation increases successful trade and welfare in the validation setting. Property-right opportunities are present, although the welfare effect is small in this medium run. The full n=20 v1 run confirms the activation pattern: successful trade appears under no institution, price controls bind and suppress trade, property-right opportunities are measured, and reputation produces the strongest welfare gains in this pressure setup.

5.4 Cross-Mind Resource Island

The Resource Island Phase 3 full table gives a different pattern from Pricing Arena. Neural/MARL minds find higher survival and welfare policies, but they do not learn the intended trade economy under the current reward and training horizon. On the no-institution condition, Q-learning has survival 0.969, welfare 1.005, and trade count 0.294. DQN has survival 0.987 and welfare 1.257 but trade count 0.000. PPO and centralized-critic show the same qualitative pattern.

This should not be over-interpreted as proof that stronger agents strategically avoid trade. The trade-attempt diagnostics suggest exploration and coordination failure: DQN and centralized-critic make rare attempts, PPO makes essentially no attempts, and attempts tend to be inventory-blocked. The safe claim is that these stronger learners find high-survival policies without discovering successful trade under the current Resource Island setup.

[TODO: Add a compact Resource Island cross-mind table with Q-learning, DQN, PPO, fixed independent-DQN, and centralized-critic on the no-institution row.]

6 Environment 3: Auction House

6.1 World

Auction House is a single-item sealed-bid auction world with independent private values, discrete bids, deterministic tie-breaking, first-price and second-price payment rules, and optional reserve prices. Each bidder observes its own valuation bin and chooses a bid. The world reports seller revenue, bidder surplus, total welfare, allocative efficiency, regret, overbidding, underbidding, and distance from benchmark bid functions.

6.2 Benchmarks

The second-price benchmark is truthful bidding. The first-price benchmark is bid shading under symmetric independent private values, with discrete-grid regret checks used where the continuous benchmark is too coarse. Reserve-price variants evaluate the revenue-efficiency tradeoff against the no-reserve benchmark.

6.3 Validation

The Auction House validation run is economically coherent but not yet final evidence. Second-price learning moves toward the truthful benchmark, first-price learning shows underbidding and bid shading, and reserve prices increase revenue while changing allocative efficiency. Table 3 summarizes the current validation run.

Scenario	Revenue	Welfare	Efficiency	Underbid Rate
second price	3.373	6.216	0.747	0.381
first price	3.481	6.331	0.768	0.774
second price + reserve	4.031	6.098	0.814	0.366

Table 3: Auction House validation results.

[TODO: Replace or extend this table with the active full P.6 Auction House ladder runs once DQN, PPO, fixed independent-DQN, and centralized-critic validate. Add learned bid curves against truthful and first-price shaded reference curves.]

7 Environment 4: Public Goods

7.1 World

Public Goods is a repeated common-pool resource environment. Agents choose between contributing to the pool, extracting from it, or remaining inactive. Extraction pays private reward but depletes the shared stock; contribution costs the contributor but helps sustain the pool. When requested extraction exceeds available stock, extraction is rationed proportionally. Deterministic regeneration makes the sustainability channel easy to audit.

The implemented institutions are no intervention, a penalty schedule, contribution matching, reputation rewards, information restriction, and a generic tax-and-redistribution schedule. The main metrics are welfare, sustainability, total contribution, extraction relative to regeneration, inequality, collapse diagnostics, and tax revenue where applicable.

7.2 Q-Learning Full Run

The n=20 tabular Q-learning full run confirms the intended commons pressure. In the baseline, agents mostly extract and contribute little: sustainability is 0.0892 and total contribution is 0.1418. Contribution matching improves both welfare and sustainability in this setting, reaching welfare 2.0808 and sustainability 0.1048. Reputation is a strong reward-shaping intervention, with welfare 9.5750, while the tax schedule mainly changes redistribution/accounting at the tested rates.

The effect validator classifies the tax schedule as mostly reward/accounting under this configuration, while penalty, contribution matching, reputation, and information restriction change state metrics relative to baseline. This is a diagnostic classification, not yet a final welfare ranking.

7.3 Cross-Mind Status

Public Goods is P.6 smoke-validated and currently running full P.6 ladder jobs. The question for the full table is whether stronger learners worsen commons collapse by extracting more efficiently, or improve sustainability by discovering contribution and institution-mediated cooperation.

[**TODO: Insert the full Public Goods P.6 comparison after the active remote queue validates.**]

8 Environment 5: Labor Market

8.1 World

Labor Market is a two-sided matching environment. Workers are learning agents; employers have fixed preferences. Each worker reports a top choice, the world constructs reported preferences, and matching is resolved using deferred acceptance. The world reports match rate, total welfare, truthful-report rate, stability, blocking-pair diagnostics, and manipulation-gain diagnostics.

This world is intentionally asymmetric. Unlike Pricing Arena, Resource Island, Public Goods, and Auction House, not every economic actor is a learning agent. That asymmetry is the main interface test for the capability ladder: DQN, PPO, independent-DQN, and centralized-critic operate only over the worker side while employers remain fixed preference holders.

8.2 Q-Learning Full Run

The n=20 tabular Q-learning full run is substantially more stable than the short smoke run. Match rate remains `1.0`, while `stability_mean=0.9508`, `truthful_report_rate_mean=0.7859`, and `total_welfare_mean=3.6393`. The report-top action space creates truthfulness variation, but worker-proposing deferred acceptance still produces mostly stable matchings under the current random-preference setup.

Fixed benchmark cases clarify the interpretation. Under worker-proposing deferred acceptance, workers should not have profitable report deviations in the canonical strategy-proof setting. Future manipulation tests therefore need to target the non-proposing side, change the mechanism, or add information and commitment frictions rather than treating worker misreports under standard deferred acceptance as the main expected failure mode.

8.3 Cross-Mind Status

Labor Market is P.6 smoke-validated and currently running full P.6 ladder jobs. The active full runs test whether stronger learners change truthful-report rates, stability, and welfare in the asymmetric worker-only learning setup.

[TODO: Insert the full Labor Market P.6 comparison after the active remote queue validates.]

9 Cross-World Synthesis

The emerging thesis is not that one institution always works. The emerging thesis is that institutional robustness must be evaluated jointly with learner capability and with activation diagnostics.

The Pricing Arena result shows capability-sensitive institutional effects: price caps reduce exploitability broadly, but stronger learners can preserve profit under the cap through a channel that a price-only collusion metric would miss. Resource Island shows the complementary failure mode: if the world design does not put an institution under pressure, then the absence of an effect is not a mechanism result. Auction House adds theory-anchored bidding benchmarks. Public Goods adds a sustainability and commons-collapse channel. Labor Market adds a matching-stability and truthfulness channel.

The preliminary five-world synthesis scaffold already exists, but three of its worlds currently use smoke-scale neural/MARL rows. Those rows are useful for checking the pipeline, not for final claims. The active full P.6 queues are therefore the current paper blocker: once Auction House, Public Goods, and Labor Market validate, the synthesis can be rewritten around five full capability-ladder tables instead of two full tables plus three smoke gates.

[TODO: After current full P.6 queues land, write this section as claims rather than a roadmap. Required inputs: final Pricing Arena table, final Resource Island table, final Auction House P.6 table, final Public Goods P.6 table, final Labor Market P.6 table, and regenerated cross-world synthesis outputs.]

10 Limitations

Several limitations are deliberate. The environments are small and stylized. They are designed for auditability rather than realism. The deep-RL agents are structured-observation MLP agents rather than large-scale policies. The Resource Island trade protocol is still simple, and the current v1 pressure configuration is an activation test rather than a realistic market. Auction House currently has benchmarked sealed-bid full runs and smoke-tested clock and information variants. Public Goods is still a stylized common-pool model rather than an empirical commons. Labor Market currently uses worker-side learning under deferred acceptance, so it tests one asymmetric matching channel rather than matching-market design generally.

The most important limitation is interpretive. A clean CSV does not imply a clean economic result. Every institution must be checked for activation. Every capability comparison must be checked for implementation artifacts. The independent-DQN alias issue is a concrete example: the table was statistically valid as an output file, but not valid as evidence for a distinct mind until the implementation was corrected and rerun.

11 Reproducibility

The repository records validation status in `ROADMAP_STATUS.md`. Core outputs are stored under `outputs/`. Major runners include:

- `run_multiseed.py` for Pricing Arena replicated runs;

- `run_exploitability.py` for frozen-policy adversarial evaluation;
- `run_phase3_full.py` and `validate_phase3_full.py` for full Phase 3 Pricing Arena checks;
- `run_resource_island_smoke.py` for Resource Island smoke, validation, and full runs;
- `run_auction_house_smoke.py` for Auction House runs;
- `run_public_goods_smoke.py` for Public Goods runs;
- `run_labor_market_smoke.py` for Labor Market runs;
- `build_world_mind_comparison.py` and `build_cross_world_synthesis.py` for ladder comparisons and cross-world synthesis.

The test suite currently includes unit, integration, and output-validation coverage for the core abstractions, Pricing Arena, Resource Island, Auction House, Public Goods, Labor Market, and deep-RL/MARL minds.

12 Conclusion

This draft reports a first version of an artificial-economies platform for testing institutional robustness under learning agents. The mature Pricing Arena result already shows why the question matters: exploitability and collusion can move differently under the same institution, and stronger learners can expose channels that weaker learners do not. Resource Island, Auction House, Public Goods, and Labor Market extend the project beyond pricing games. Their role is not merely to add more checkboxes, but to test whether the patterns found in Pricing Arena survive when the economic structure changes.

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